

WATER CONSUMPTION FORECASTING & ANOMALY DETECTION SYSTEM

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: Water Consumption Forecasting & Anomaly Detection System

Technology Stack:

Python | Machine Learning | Scikit-Learn | RandomForest | Prophet | Pandas | NumPy | Streamlit | Plotly

Project Description:

An end-to-end machine learning system developed to forecast household and district-level water consumption, identify unusual consumption spikes/drops, and categorize demand. The project uses machine learning models (Random Forest, Prophet, etc.) for demand prediction, statistical anomaly detection (rolling Z-Score), and features an interactive Streamlit dashboard with real-time metric tiles and dynamic data exports to visualize insights and support water resource management.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission: 02/08/2026

1. Project Objective

The objective of this project is to develop a machine learning system that forecasts household and district-level water consumption using historical water usage records, seasonal patterns, and temporal trends.

The system predicts future water demand, identifies unusual consumption spikes/drops, and provides a fully interactive UI. This enables utility providers to optimize water distribution, perform resource planning, and prevent infrastructure damage (e.g., detecting pipe leaks via consumption spikes).

The project includes:

- Exploratory Data Analysis
- Data preprocessing
- Feature engineering (Lags & Rolling windows)
- Multi-algorithm demand forecasting
- Model comparison
- Demand categorization (Low/Medium/High)
- Anomaly Detection (Spike/Drop Alerts)
- Interactive Streamlit dashboard

2. Requirement Coverage

Requirement	Status	Project File / Output
Consumption Forecasting Model	Completed	models/random_forest_model.pkl
Feature Engineering	Completed	notebooks/02_feature_engineering.ipynb
Cleaned Dataset	Completed	Dataset/processed/
Anomaly Detection (Spike/Drop)	Completed	src/anomaly_detector.py
Demand Categorization	Completed	src/anomaly_detector.py
EDA Report	Completed	reports/EDA_Report.md
Model Evaluation	Completed	reports/Forecast_Evaluation_Report.md
Streamlit Dashboard	Completed	Dashboard/app.py
Project Documentation	Completed	Documentation/Final_Documentation_Report.docx

3. Dataset

The project uses water consumption historical datasets:

Features:

- date (Temporal index)
- consumption_liters (Target Variable)
- region (Categorical)

4. Work Completed

Completed an end-to-end AI/ML workflow including:

- Loaded and inspected historical water consumption datasets.
- Performed exploratory data analysis and visual distribution mapping.
- Created advanced temporal and rolling lag features.
- Trained 7 distinct machine learning algorithms (Prophet, SARIMA, LSTM, GRU, Random Forest, LightGBM, XGBoost).
- Compared model performance and evaluated metrics (RMSE, MAE, MAPE).
- Selected Random Forest for dashboard deployment.
- Implemented Z-Score based statistical anomaly detection engine.
- Developed a fully functional, multi-page Streamlit dashboard with glassmorphism aesthetics.
- Exported processed datasets, detailed markdown reports, and trained models.

5. Methodology

5.1 Exploratory Data Analysis

Performed analysis on:

- Consumption distribution
- Region-wise boxplots
- Day of Week Trends

5.2 Feature Engineering

Created additional analytical features:

- 1-day, 7-day Lag Features
- Rolling Mean (7-day)
- Rolling Std Dev (7-day)
- Temporal features (day_of_week, is_weekend, month, day)

6. Model Development

Multiple regression models were trained and compared:

- Prophet
- SARIMA
- LSTM
- GRU
- Random Forest Regressor
- LightGBM
- XGBoost

The models were evaluated using MAE, RMSE, MAPE, and R2 Score.

7. Model Results

Model	MAE	RMSE	MAPE (%)
Prophet	1636.42	2047.43	14.19
SARIMA	1793.68	2363.29	16.22
Random Forest	3455.92	4128.54	34.99
LightGBM	3645.82	4423.02	35.62
XGBoost	3728.25	4580.85	37.73

8. Selected Model

Random Forest Regressor

Reason for selection:
While statistical models performed best in a vacuum on the test set, Random Forest provided the best overall performance among the machine learning algorithms that could be easily deployed for instant, row-by-row tabular inference in the Streamlit web app without requiring complete sequential index context.

9. Streamlit Application

The Streamlit application is implemented in:
Dashboard/app.py

The dashboard contains:

Executive Dashboard & Forecast Overlay

Displays KPIs, current status, interactive Plotly charts overlaying actual vs AI-forecasted consumption, and active anomaly alerts.

EDA & Analytics Page

Displays statistical distribution plots and region-wise consumption mapping.

Model Evaluation Page

Displays performance metric comparisons across all 7 evaluated algorithms.

10. Key Findings

- Identified cyclical patterns in water consumption accurately using time-series models (Prophet/SARIMA).
- Successfully deployed tree-based models for scalable real-time inference.
- Detected critical demand spikes and structural drops using statistical Z-score rolling windows.

11. Recommendations

- Incorporate external weather regressors (temperature, rainfall) to improve prediction accuracy.
- Expand neural network (LSTM/GRU) look-back windows for deeper context learning.
- Establish automated alerting pipelines when severe "Spike" anomalies are detected.

12. Achievements

Successfully completed:

- Data preprocessing & Exploratory analysis
- Machine learning model development and comprehensive evaluation (7 models)
- Water consumption anomaly detection engine
- End-to-End multi-page Streamlit Dashboard deployment with premium aesthetics

13. Limitations

- Dataset lacks exogenous meteorological indicators (weather) which heavily drive human water consumption.

14. Final Submission Contents

- Documentation Report (.docx)
- Machine Learning Models (.pkl)
- Streamlit Dashboard (app.py and pages/)
- Processed Datasets
- Markdown Reports
- Jupyter Notebooks

15. How to Run

```
pip install -r requirements.txt
```

```
streamlit run Dashboard/app.py
```

16. Final Conclusion

The Water Consumption Forecasting & Anomaly Detection System successfully delivers an end-to-end machine learning solution for demand forecasting, statistical outlier monitoring, and interactive visualization.

The system combines predictive AI with robust real-time anomaly alerts to help utility managers optimize water distribution, identify leaks, and make data-driven decisions through a fully interactive, premium Streamlit dashboard.